Causal-Temporal Machine Learning for Root-Cause Localization in 5G Networks: A Benchmark Audit and a Cautionary Evaluation

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Abstract

We audit TelecomTS and find that our initial headline does not survive correction of the benchmark. We repair a train/test background leak and injection-label errors, expose a per-fault-type template shortcut, add instance-varying injections that test it, and retrain on the repaired split (5,689 test windows, 424 positives, 26 recordings). The prior result collapses. TAGN, our temporal-attention graph network, reaches 0.7589 PR-AUC across five seeds, below TCN (0.8606) and logistic regression (0.8196). Removing the graph raises PR-AUC from 0.7439 to 0.7581: the graph is not a marginal contributor, it is worse than no graph in a single-seed ablation. The learned graph estimates statistical dependencies; it is not a verified causal graph. The intervention-based RCA metric fails its own positive control: the ground-truth root set scores 0.496 ± 0.052, essentially the random-root null at 0.503 ± 0.016; the full true set against a matched random set gives 0.508 ± 0.085. The predicted-root score (0.567 ± 0.235) is unstable across seeds, ranging from 0.304 to 0.976, not evidence of localization. The old argmax-lag early-warning number is also an artifact: under a calibrated event-level protocol, detection at a 1% false-alarm budget is 3.7%, with per-step AUC 0.495. We claim no state-of-the-art model. We claim a corrected benchmark, a validation protocol that separates real RCA from benchmark artifacts, and a decisive negative result: the graph provides no clear detection benefit in the ablation, the causal interpretation fails validation, and our initial headline is unsupported.

1 Audit and Repairs

We audited TelecomTS along four axes: split integrity, preprocessing leakage, ground-truth validity, and metric protocol. Three findings changed the numbers: (G5) a train/test background leak in which a test anomaly's normal source was chunk-split into train; (G2–G4) injection label errors (identity/inert effects labeled, multi-event windows under-labeled, `end_time` 10× too large); and a template shortcut in which the affected-KPI set is constant per fault type. An input-free type-oracle reaches 0.971 top-k Jaccard on the shipped data, above every trained model; after instance-varying injection the ceiling drops to 0.661. We bind sources atomically to their injected twins, regenerate the corpus (280 spurious labels removed across 237/714 windows), and refit scalers and graph prototypes per leave-recordings-out fold. The repaired test set has n = 5,689, 424 positives, and 26 recordings. The repairs are not cosmetic. They reverse the headline.

2 Detection

Model	PR-AUC	ROC-AUC
TCN	**0.8606**	0.9438
LogReg	0.8196	0.9309
TAGN (graph + summary, 5 seeds)	0.7589	**0.9543**
GRU	0.7635	0.9399

Table 1: Setting I, repaired split, recording-level cluster-bootstrap protocol.

TAGN does not win. It trails TCN and logistic regression on mean PR-AUC. Its best seed (1337, 0.8278) edges logistic regression but remains below TCN. Paired cluster-bootstrap tests against LogReg are negative on 4/5 seeds and every 95% interval includes zero: no advantage is detectable, and TAGN loses on the point estimate. The single-seed component ablation is also negative. Removing the graph gives 0.7581 PR versus 0.7439 for the full model, so the graph is not neutral; no_graph is the best variant. Message passing contributes nothing beyond summary statistics.

3 Interventional Fidelity Fails Its Positive Control

We define a model-internal do-style perturbation by zeroing selected normalized KPI channels and re-scoring the anomaly head. The fidelity is the fraction of windows in which intervening on a candidate root set reduces the score more than intervening on a matched random non-root set. A valid metric must pass a positive control on the ground-truth affected-KPI set. It does not: across five checkpoints and 424 anomalous test windows from 13 recordings, true roots score 0.496 ± 0.052 at $k = 3$, 0.476 ± 0.086 at $k = 5$, and 0.519 ± 0.137 at $k = 7$, while random roots score 0.503 ± 0.016, 0.506 ± 0.026, and 0.495 ± 0.025. The full true set against a matched random set gives 0.508 ± 0.085, so truncation is not the reason for the failure. This is not a borderline result: the ground-truth-root score is at chance and overlaps the matched random-root null. The predicted-root score is 0.567 ± 0.235, but it is unstable across seeds (0.304 to 0.976); one checkpoint drives the mean. Because the positive control fails, we do not interpret this as evidence of causal localization. The earlier pilot value of 0.465 is invalid: it used a single checkpoint without matched controls and is not reproducible under the corrected implementation. The learned graph is a statistical dependency graph, not a verified causal DAG; the intervention is a model-internal perturbation, not a physical do-operation.

4 Early Warning

The usual argmax-lag score reports 65.2% "warning before onset," but the argmax can occur after onset and is not calibrated to a false-alarm budget. We instead choose a threshold on validation normal windows and require the first pre-onset alarm, then report event-level detection and false alarms. Over five seeds and 93 test events, a 1% false-alarm budget yields 3.7% detection (0.018 false alarms per normal window); at 5% it yields 10.1% detection (0.096 false alarms per window). Per-step AUC is 0.495 and AP is 0.006. The old number is not an early-warning result. Under a calibrated protocol the hazard head is near chance and is not usable as an early-warning system on this data.

5 Conclusion

The audit changes the conclusions. After repair, TAGN trails TCN and logistic regression on mean PR-AUC, the single-seed graph ablation favors no graph, the intervention metric fails its positive control, and the calibrated hazard head is near chance. The learned graph is a dependency estimate, not a causal mechanism. We put intervention-based RCA on trial and it failed; we replaced the broken early-warning metric and the head failed there too. These are decisive audit findings. We release the pipeline, the repaired split, the validation scripts, and the test suite so that others can reproduce the audit and apply the same controls to their own claims. The initial headline does not hold under the repaired benchmark and protocol.